Chapter 1 · Energy Causes Change

Name: _____ **Homeroom:** ◯ 4A ◯ 4B ◯ 4C ◯ 4D

MONDAY · BQ / Vocabulary / Evidence

TEACHER KEY

BQ Big Question

Where can we observe evidence of energy causing change?

My first thinking Write 1–2 quick ideas. No wrong answer yet.

We can observe evidence of energy when a pencil gets shorter and sharper in the electric sharpener.

Sample answer — student wording may vary.

V1 Vocabulary — Energy & Change

energy — the ability to cause ***change*** .

change — to become ***different*** .

N1 Energy Causes Change

Energy has the ability to cause change.

A change is ***evidence*** that energy was involved.

The Chapter 1 model:



Reader pp. 1–2

Chapter 1 · Energy Causes Change

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TUESDAY · Forms / Stored Energy

TEACHER KEY

N2 Energy occurs in different forms

Write one example that shows a change for each form.

FORM	ONE EXAMPLE THAT SHOWS A CHANGE
motion	*a ball rolls across the floor*
sound	*a drum makes a sound*
light	*a flashlight lights a dark area*
heat	*hands become warmer near a warm object*
electrical energy	*a fan begins to spin when plugged in*

Sample answers — student wording may vary.

Reader p. 3

V2 Vocabulary — Stored Energy

stored energy — energy that has the ability to cause change at a ***later*** time.

Reader p. 4

N3 Objects can store energy

STORED ENERGY → **RELEASE** → **CHANGE**

EXAMPLE	WHAT CHANGE COULD HAPPEN LATER?
battery (in a flashlight)	*the flashlight can light up*
stretched rubber band	*the band can snap back and move*
raised object (apple)	*the apple can fall*

Sample answers — student wording may vary.

Reader pp. 4–5

Objects can also have stored energy because of their ***position***.

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FRIDAY · Evidence / Reflection

TEACHER KEY

N4 Recognizing Evidence

Draw a change you observed. Show what it looked like BEFORE and AFTER.

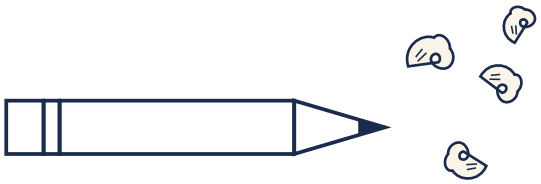
Reader pp. 1–5

BEFORE *Sample drawing*



long pencil, dull rounded tip

AFTER *Sample drawing*



shorter, sharp point

What changed? *The pencil became shorter and sharper.*

Sample answer — student wording may vary.

CHECK Your Thinking

How do we know energy is involved if we cannot see energy itself?

We know because we see ... ***a change, like the pencil getting shorter. A change is evidence of energy.***

Sample answer — student wording may vary.

BQ Big Question — Now I think

Where can we observe evidence of energy causing change?

We can observe evidence of energy when something changes, like a pencil getting sharper, a drum making sound, or a flashlight lighting a dark room.

Sample answer — student wording may vary.